



UNITED INTERNATIONAL UNIVERSITY
Department of Computer Science and Engineering (CSE)
Course Syllabus

1	Course Title	Electronics Laboratory
2	Course Code	EEE 2124
3	Trimester and Year	Summer 2024
4	Pre-requisites	EEE 2113
5	Credit Hours	1.00
6	Section	E
7	Class Hours	Sunday : 11:11:AM - 01:40:PM
8	Class Room	Circuit Lab 1 (Room: 505)
9	Name of the Faculty	Rifat Bin Rashid Lecturer, Dept. of CSE
10	Contact	Email: rifatbinrashid@cse.uiu.ac.bd
11	Office	Room# 219, Permanent Campus
12	Counselling Hours	
13	Teaching Assistant	N/A
14	Text Book	None.
15	Reference	1. Electronic devices and circuit theory (BOYLESTAD & NASHELSKY)-11th edition 2. Microelectronic Circuits (Sedra / Smith)-5th edition 3. Arduino Project Handbook: 25 Practical Projects to Get You Started by Mark Geddes 4. Programming the Raspberry Pi: Getting Started with Python by Simon Monk
16	Course Contents (approved by UGC)	Electronics laboratory explores the design, construction, and debugging of electronic circuits. Lectures and laboratory tasks investigate the performance characteristics of diodes, transistors, MOSFETs and op-amps. The design and implementation of a project in a team is a must. Provides opportunity to simulate circuits both in hardware and software. Students learn about the basic principles of circuit design and operation in a practical, real-world laboratory setting. They work both with discrete components such as resistors, capacitors, diodes, and transistors as well as with integrated components such as operational amplifiers. In addition, they become familiar with the operation of basic electronic test equipment (digital multi-meters, oscilloscopes, function generators etc). Electronics laboratory helps the students to develop a complete project of their own, which includes both of hardware and software design and developments.

17	Course Outcomes (COs)																																																			
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C01	Utilize basic electrical and electronic components, power supply, signal generators and measurement devices to implement simple circuits on breadboard.																																																			
C02	Analyze electronic circuits using computer simulation software to compare experimental results with computer-aided analysis.																																																			
C03	Design a simple electronic system as a course project based on lessons learned in EEE 2123.																																																			
C04	Prepare a written report for each experiment to effectively communicate the experimental data and lessons learned at the laboratory session.																																																			
C05	Exhibit the ability to work in a team to perform laboratory experiments and project.																																																			
18	Teaching Methods	Lecture, Laboratory exercise and reports, Project Development.																																																		
19	CO with Assessment Methods	<table><tr><td>CO</td><td>Assessment Method</td><td>University prescribed(%)</td><td>Actual weight(%)</td></tr><tr><td>-</td><td>Attendance</td><td>10</td><td>10</td></tr><tr><td>4</td><td>Class Performance</td><td>20</td><td>-</td></tr><tr><td>3</td><td>Report</td><td>10</td><td>10</td></tr><tr><td>2</td><td>Project</td><td>20</td><td>40</td></tr><tr><td></td><td>Mid Term</td><td>15</td><td>20</td></tr><tr><td>1</td><td>Final Exam</td><td>25</td><td>20</td></tr></table>	CO	Assessment Method	University prescribed(%)	Actual weight(%)	-	Attendance	10	10	4	Class Performance	20	-	3	Report	10	10	2	Project	20	40		Mid Term	15	20	1	Final Exam	25	20																						
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	10	2	PROJECT UPDATE		Hardware implementation
	11	2	PROJECT UPDATE		Hardware implementation
	12	2	PROJECT SHOWCASE		Hardware implementation

Appendix 1: Assessment Methods

Assessment Types	Marks
Attendance	10%
Class Performance	20%
Report/Viva	10%
Presentation(on Project)	20%
Mid Term	15%
Final Exam	25%

Appendix 2: Grading Policy

Letter Grade	Marks %	Grade Point	Letter Grade	Marks %	Grade Point
A (Plain)	90-100	4.00	C+ (Plus)	70-73	2.33
A- (Minus)	86-89	3.67	C (Plain)	66-69	2.00
B+ (Plus)	82-85	3.33	C- (Minus)	62-65	1.67
B (Plain)	78-81	3.00	D+ (Plus)	58-61	1.33
B- (Minus)	74-77	2.67	D (Plain)	55-57	1.00
			F (Fail)	<55	0.00

Appendix-3: Program outcomes

POs	Program Outcomes
PO1	An ability to apply knowledge of mathematics, science, and engineering
PO2	An ability to identify, formulate, and solve complex engineering problems
PO3	An ability to design solutions for complex engineering problems and design systems, components or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations
PO4	An ability to investigate complex problems using research-based knowledge and research methods design and conduct experiments, as well as to analyze and interpret data
PO5	An ability to use the techniques, skills, and modern engineering tools necessary for engineering practice
PO6	The broad education necessary to understand the impact of engineering solutions in a global, economic, environmental, and societal context
PO7	Understand and evaluate the sustainability and impact of professional engineering work in the solution of complex engineering problems in societal and environmental contexts
PO8	An understanding of professional and ethical responsibility
PO9	An ability function effectively as an individual, and as a member or leader in diverse teams and in multi-disciplinary settings
PO10	An ability to communicate effectively
PO11	Project management and finance
PO12	A recognition of the need for, and an ability to engage in life-long learning